Problems: Foundations

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Complex Numbers

$|c|$ - the magnitude of complex number c
$\arg\{c\}$ - the phase of the complex number c
c^* - the complex conjugate of c $\mathrm{Re}\{c\}$, $\mathrm{Real}\{c\}$ - the real part of c
$\mathrm{Im}\{c\}$, $\mathrm{Imag}\{c\}$ - the imaginary part of c

The complex number c may be written in rectangular coordinates as $c = \mathrm{Re}\{c\} + j\mathrm{Im}\{c\}$, and in polar coordinates as $c = |c|e^{j \arg\{c\}}$.

1. A complex number $a = 3 - 4j$ where j is the square root of -1. Find:

   a) $|a|$

   b) $\arg\{a\}$ in radians

   c) $\mathrm{Imag}\{a^*\}$

   d) aa^*

2. Let $a = -5$. Find:

   a) $|a|$

   b) $\arg\{a\}$ in radians

3. Let $a = -4j$. Find:

   a) $|a|$

   b) $\arg\{a\}$ in radians

4. A complex function $a(t) = 2t - j2t^2$. Find $\arg\{a(t)\}$ in radians.

5. Let $a(t) = 5te^{j3t}$ and assume $t > 0$. What is $\mathrm{Re}\{a(t)\}$? Select all correct answers.

   a) $5t$

   b) $3t$

   c) $5t\cos(3t)$

   d) $5t\sin(3t)$

   e) $5t\tan^{-1}(3t)$

   f) $0.5a(t) + 0.5a^*(t)$

   g) $0.5a(t) - 0.5a^*(t)$

   h) $\sqrt{a(t)a^*(t)}$

6. Let $a(t) = e^{-t^2}e^{j2t}$ and $b(t) = |t|e^{j3t+\pi/2}$. Find $\arg\{c(t)\}$ if $c(t) = a(t)b(t)$. Select all correct answers.

a) $-t^2 + 5t + \pi/2$

b) $5t + \pi/2$

c) $\pi/2$

d) $|t|e^{-t^2}$

e) $-t - \pi/2$

f) $\tan^{-1}\left\{\frac{3t+\pi/2}{2t}\right\}$

g) $\tan^{-1}\left\{\frac{\text{Im}\{b(t)\}}{\text{Re}\{a(t)\}}\right\}$

h) $t + \pi/2$

7. Write $x(t) = (1 + 2j)e^{j3t} + (1 - 2j)e^{-j3t}$ using real-valued sinusoids.

a) $x(t) = \cos(3t) + 2\sin(3t)$

b) $x(t) = 2\cos(3t) + 4\sin(3t)$

c) $x(t) = \cos(3t) - 2\sin(3t)$

d) $x(t) = 2\cos(3t) - 4\sin(3t)$

e) $x(t) = 2\cos(3t) - 4j\sin(3t)$

f) $x(t) = 2\cos(3t) - 4\sin(6t)$

Discrete- and Continuous-Time Frequency

8. A discrete-time sinusoid $x[n] = 3\cos(\pi n/5 + \pi/4)$ is obtained by sampling a continuous-time sinusoid $x(t)$ at sampling interval $T = 0.01$ s. That is, $x[n] = x(nT)$. Which of the following choices could have been the frequency of $x(t)$? Select all that are correct.

a) 100 Hz

b) 5 Hz

c) 10 Hz

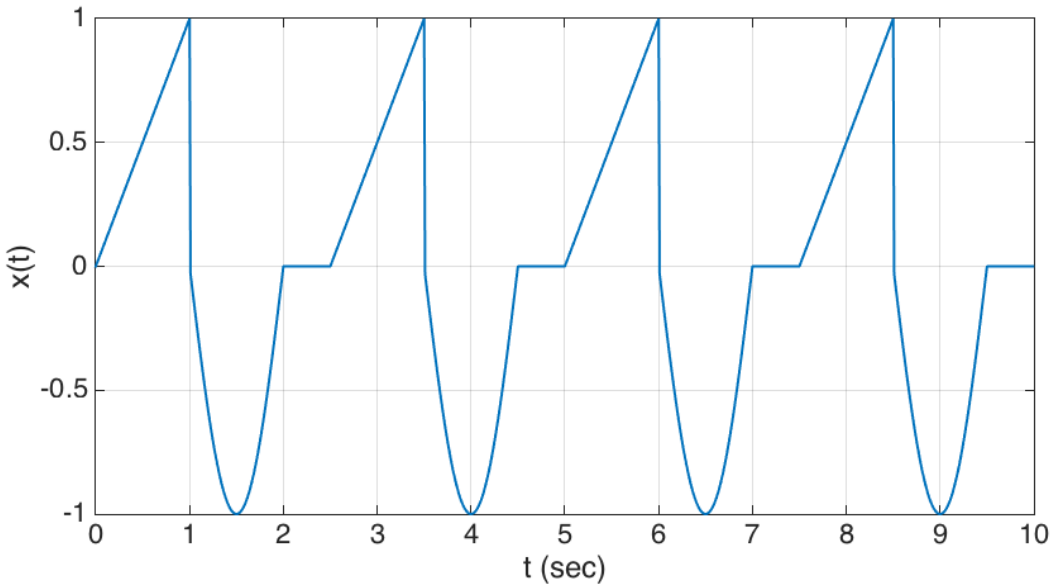
d) 110 Hz

e) 90 Hz

f) 55 Hz

Periodic Signals

9. Find the fundamental period in sec and fundamental frequency in Hz of the periodic continuous-time signal depicted below.



10. If a signal $x(t)$ has fundamental period $T_x = 4$ s and $y(t)$ has fundamental period $T_y = 5$ s, what is the fundamental period of $x(t) + y(t)$?

11. If a signal $x(t)$ has period T_x s and $y(t)$ has period T_y s, what condition must be satisfied for $x(t) + y(t)$ to also be periodic? Assume m and k are both positive integers.

a) $T_x = T_y + m$

b) $T_x = mT_y$

c) $T_y = kT_x$

d) $mT_y = kT_x$

e) $T_xT_y = mk$

f) $T_x + T_y = m + k$

System Properties

12. The output $y(t)$ of a system is obtained by delaying the input by 2 secs, multiplying by three, and changing its sign. That is, $y(t) = -3x(t-2)$.

a) Is this system time invariant?

b) Is this system memoryless?

c) Is this system BIBO stable?

d) Is this system linear?

e) What is the impulse response $h(t)$ of the system?

i) can't be determined from information given

ii) $h(t) = -3u(t-2)$

iii) $h(t) = 3\delta(t-2)$

iv) $h(t) = 3u(t-2)$

v) $h(t) = -3\delta(t-2)$

vi) $h(t) = -3\delta(t+2)$

vii) $h(t) = 3\delta(t+2)$

Impulse Response Descriptions

13. An LTI system has impulse response $h[n] = (1/2)^n u[n]$.

a) The input to the system is $x[n] = 2\delta[n-3]$. Use linearity and time invariance to find the output.

b) The input to the system is $x[n] = 4\delta[n] - 2\delta[n-3]$. Use linearity and time invariance to find the output.